THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET B

Chapters 7 & 8: **The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An event occurred 46 times in 200 trials. Its experimental probability (relative frequency) is: 1

(a) 4.35 **(b)** 0.23 **(c)** 0.33 **(d)** 0.115

2. The number of outcomes in the sample space when two coins are tossed together is: 1

(a) 4 **(b)** 8 **(c)** 6 **(d)** 2

3. A sample shows a probability of $\frac{3}{10}$ that a student likes cricket. In a school of 2000 students the expected number who like cricket is: 1

(a) 800 **(b)** 700 **(c)** 300 **(d)** 600

4. The probability of a certain event is: 1

(a) 0 **(b)** 100 **(c)** 1 **(d)** 0.5

5. As the number of trials increases, the experimental probability tends to get closer to the: 1

(a) value 1 **(b)** theoretical probability

(c) sample space size **(d)** number of trials

6. The 6th term of the GP with first term 3 and common ratio 2 is: 1

(a) 97 **(b)** 95 **(c)** 192 **(d)** 96

7. The 7th term of the GP with first term 1 and common ratio -2 is: 1

(a) 65 **(b)** 63 **(c)** 64 **(d)** 128

8. If $T_1 = 1, T_2 = 2, T_3 = 4$ and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$, then T_5 is: 1

(a) 11 **(b)** 13 **(c)** 12 **(d)** 14

9. The sequence $1, 3, 6, 10, 15, \ldots$ (triangular numbers) is: 1

(a) neither an AP nor a GP **(b)** an AP with d = 5

(c) an AP with d = 2 **(d)** a GP with r = 3

10. The number of black triangles at Stage n of the Sierpiński triangle is: 1

(a) $3n$ **(b)** 4^n **(c)** 2^n **(d)** 3^n

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A box has 20 bulbs of which 3 are defective. One bulb is picked at random. Find the probability that it is 2

(i) defective

(ii) not defective.

12. Two dice are rolled together. Find the probability that the sum is 7. 2

13. Find the nth term of the AP $11, 8, 5, 2, \ldots$ and write its recursive rule. 2

14. Find the sum of the first 20 natural numbers and of the first 100 natural numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A number is chosen at random from 1 to 25. Find the probability that it is 3
(i) a multiple of 5
(ii) a prime number.

16. A bacterial colony of 500 doubles every hour. How many bacteria are there after 6 hours? 3

17. A basketball player scored 18 times in 30 attempts. Find the experimental probability of scoring and estimate the number of scores in 50 attempts. 3

OR

Explain what happens to experimental probability as the number of trials increases, using the Law of Large Numbers. 3

18. Write the first ten terms of the Virahānka–Fibonacci sequence ($V_1 = 1, V_2 = 2$). 3

OR

Find the first five terms of $u_1 = 1,\ u_n = 2u_{n-1} + 3$ and decide whether 133 is a term. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain subjective probability with an example. 5
(ii) Explain how an objective estimate of the chance of rain can be made from data.
(iii) Why is rain called a random event?

OR

Case study: Two players A and B play with a spinner of 8 equal sectors numbered 1–8. A wins if the number is prime; B wins if the number is a composite number. 5
(i) Find $P(A \text{ wins})$.
(ii) Find $P(B \text{ wins})$.
(iii) Is the game fair?
(iv) Which number(s) make neither win?

20. (i) Plot the points (n, t_n) for the AP $1, 5, 9, 13, 17$ and say what you observe. 5
(ii) Plot the points for the GP $3, 6, 12, 24, 48$ and compare.
(iii) What does this tell you about growth in an AP and a GP?

OR

Case study: Meera saves ₹100 in the first month and ₹50 more each month than in the previous month. 5
(i) Write her savings for the first three months.
(ii) Find the nth term.
(iii) Find her saving in the 12th month and her total in a year.
(iv) In which month does she save ₹1,000?

END OF QUESTION PAPER — **Total: 40 Marks**